

## Original article

## Parenteral analgesics for pain relief in acute pancreatitis: A systematic review

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## ABSTRACT

**Aim:** To assess the efficiency and safety of parenteral analgesics for pain relief in acute pancreatitis.**Materials and methods:** We carried out an electronic search of PubMed, Cochrane Library, EMBASE, WEIPU, CNKI and CBM and a manual search for eligible studies. The methodological quality of included trials and quality of evidence were examined by the Cochrane Collaboration's tool for assessing risk of bias and GRADE, respectively. The data were mainly analyzed descriptively and some were pooled by Review manager 5.**Results:** Eight randomized controlled trials with a total of 356 patients were finally included in this systematic review. Compared with procaine, pentazocine led to lower pain severity: day 1, Mean Difference (MD), 95%CI: 40.0 [35.3, 44.7]; day 2, MD, 95%CI: 24.00 [20.88, 27.12]; day 3, MD, 95%CI: 5.00 [2.17, 7.83], and it decreased the requirement for additional analgesics, Relative Risk, 95%CI: 2.23 [1.63, 3.05]. The combination of fentanyl, atropine, droperidol and lidocaine rendered lower pain score: day 1, MD, 95%CI: -5.46 [-6.95, -3.97]; day 2, MD, 95%CI: -5.78 [-7.39, -4.17]. Patients treated with metamizole tended to had lower pain than those treated with morphine, MD, 95%CI: -2.60 [-2.95, -2.25]. Nausea, emesis and vomiting were the common adverse events reported and there was almost no significant difference between different agents on safety.**Conclusions:** The systemic review showed that the randomized controlled trials comparing different analgesics were of low quality and did not favor clearly any particular analgesic for pain relief in acute pancreatitis.

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## 1. Introduction

Acute pancreatitis is an inflammatory condition with a variable clinical course. It has multiple etiologies, and might create a catabolic stress state [1]. There are various causes among which biliary stone migration is the most frequent one, followed by alcoholic etiology, hypertriglyceridaemia, hyperparathyroidism, pancreatic malignancy, ERCP, etc [2]. Globally, acute pancreatitis is a common disease with an incidence of approximately 30 per 100,000 persons per year [3]. Acute pancreatitis is a potential fatal disease with a high overall mortality (2.1%–7.8%) [4,5]. For the severe acute pancreatitis, mortality may be as high as 25% [6]. The current interventions for acute pancreatitis mainly involve supportive therapy, surgical and radiological interventions [7,8].

Abdominal pain is one of the most common symptoms as well as the most serious complains from acute pancreatitis patients particularly for those with the severe type [9]. Pain alleviation is a vital step in the management of acute pancreatitis because it may lead to many problems like anxiety and even respiratory distress [10,11]. The current interventions for pain alleviation of acute pancreatitis mainly include parenteral narcotics [10,11], epidural analgesia [12,13] and Chinese medicine [14,15]. Parenteral analgesics, which include buprenorphine, pethidine, morphine, fentanyl are the interventions frequently used in this setting [10,11]. Overall, the parenteral analgesics for acute pancreatitis can be divided into three categories: opioid analgesics, local anesthetic and nonsteroidal anti-inflammatory drugs (NSAIDs). Opioid analgesics, like pethidine and morphine, relieve the pain by mimicking the function of the endogenous opioid peptides which active the opioid receptor [16]. Opioid analgesics are widely applied to relieve pain of various causes like surgery and advanced cancer, but they may cause spasm in the sphincter of Oddi, addiction or tolerance, which

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should be carefully concerned. The systemic administration of local anesthetics is assumed to relieve pain via anti-inflammatory, neuroprotective, and motility-modulating effects [17,18]. The mechanism of NSAIDs is mainly via inhibiting the synthesis of prostaglandins, and NSAIDs are alternatively used to improve disease symptoms in acute pancreatitis [19]. Up to date, a number of clinical trials have been carried out to study the efficacy and safety of these parenteral analgesics, but there is still limited evidence that prove the superiority of these interventions and some studies even indicated opposite results [6,22,24]. In this study, we conducted a systematic review of current clinical studies with aim to assess the efficacy and safety of parenteral analgesics for pain relief in acute pancreatitis.

## 2. Methods

The systematic review was conducted and reported based on the recommendation of the Cochrane Collaboration [20] and PRISMA statement [21], respectively.

### 2.1. Criteria for considering studies for this review

We included randomized controlled trials (RCT) which compared different parenteral analgesics (meperidine, morphine, fentanyl, buprenorphine, procaine, pethidine, etc), or compared analgesics with placebo/no treatment for pain alleviation in acute pancreatitis. Diagnosis of acute pancreatitis was defined by the Atlanta criteria/Santorini criteria [6,22]. The primary outcomes are three folds: (1) pain scores; (2) needs for additional analgesic medication; (3) side effects satisfaction with pain management, blood pressure and laboratory parameters were considered as the secondary outcomes. Trials that using Chinese medicine and epidural analgesia for pain management in acute pancreatitis were excluded.

### 2.2. Literature search

Before the search strategy was finally formulated, we have consulted four clinical experts in the First Hospital of Lanzhou University to make sure the search strategy was comprehensive and systematic. The search strategy mainly included terms of acute pancreatitis, pain and a specific filters for retrieving RCTs. A total of six online databases were searched, which are PubMed, Cochrane Library, EMBASE, WEIPU, CNKI(National Knowledge Infrastructure) and CBM(Chinese BioMedical Literature Database). There was no language restriction and the date was limited up to June 2012. We also carried out a manual search of eight journals for those early studies which were not available from the six databases mentioned above. The journals we manually searched are: Journal of Peking University (Health Sciences)(1959–1985); The Journal of Practical Medicine (1972–1979); Medical Journal of Chinese People's Liberation Army (1964–1985); Journal of Traditional Chinese Medicine (1964–1984); Shanghai Medical Journal (1973–1985); Pain Clinic Journal (1964–1979); Chinese Journal of Pain Medicine (1995–1985). The reference lists of the original papers were checked to identify further studies.

### 2.3. Selection of studies and data extraction

Two reviewers (Jinqiu Yuan and Wenbo Meng) independently carried out the study selection and data extraction, and a cross check was performed to ensure the validity. The study selection was divided into three steps. The search results were firstly imported into Endnote and duplicated citations were removed. Then titles and abstracts were reviewed for further evaluation. Finally, full text

articles were retrieved for detailed review. As for data extraction, a standard form was designed and we pre-extracted two studies to ensure that the form was comprehensive and appropriate. The extracted data consisted study characteristics (titles, authors, interventions outcomes, etc.), study methods (including grouping, blinding, etc.) and the main results.

### 2.4. Critical appraisal and quality of evidence

Jinqiu Yuan and Wenbo Meng independently assessed the methodological quality and the quality of the evidence. The methodological quality was assessed based on the recommendation by Cochrane Handbook [20] which includes six domains: sequence generation, concealment of allocation, blinding, incomplete outcome data, selective reporting and other sources of bias. Each domain of risk of bias was graded as “yes” for low risk; “unclear” for moderate risk and “no” for high risk. Furthermore, we also evaluated the quality of the evidence on primary outcomes with GRADE [23]. According to GRADE, randomized trials and observational studies were initially considered as high and low quality evidence, respectively. The quality of evidence was downgraded if there were study limitations, inconsistency of results, indirectness of evidence, imprecision and publication bias, and upgraded for large magnitude of effects or plausible confoundings which would reduce a demonstrated effect, or dose–response gradient.

### 2.5. Data analysis

Because most comparisons involve only one study, we mainly analyzed included studies descriptively. The data about the need for additional analgesics between procaine and placebo was pooled with a fixed model (there was no significant heterogeneity). Most data about comparison effect was summarized and reanalyzed if possible. The reanalysis was conducted by Review manager. The risk of publication bias was planned to be analyzed by funnel plot if sufficient studies were included.

## 3. Results

### 3.1. Study characteristics and the quality of included studies

We initially identified 1691 citations and of which, 8 RCTs were finally included (see Fig. 1) [24–31]. Table 1 provides an overview of the baseline characteristics of the 8 trials. The number of patients included in these studies was small (ranged from 16 to 101). Most of the studies were conducted in Europe after year 2000, and procaine, pethidine and fentanyl were the main analgesics studied. Six studies were published in English, one in Chinese and one in Germany. Table 2 provides an overview of the methodology quality of included studies. We assessed the quality of evidence on the key outcomes: pain severity, pain reduction, need for additional analgesics and side effects. Overall, the risks of bias and imprecision of results were the main reasons to downgrade the quality of evidence and we hardly found any possible factor for upgrading. The main findings and the quality of evidence were summarized in Table 3.

### 3.2. Procaine vs. placebo

Two studies [24,25] compared procaine with placebo but the results were inconsistent. The data on pain severity was not able to be pooled. One study [24] showed that the pain severity was comparable throughout the study period but the other study [25] suggested that procaine was associated with a stronger decrease in visual analog scale (VAS) (median, –62 vs –39,  $P = 0.025$ ). Similarly, one study [24] suggested the need for additional analgesics was

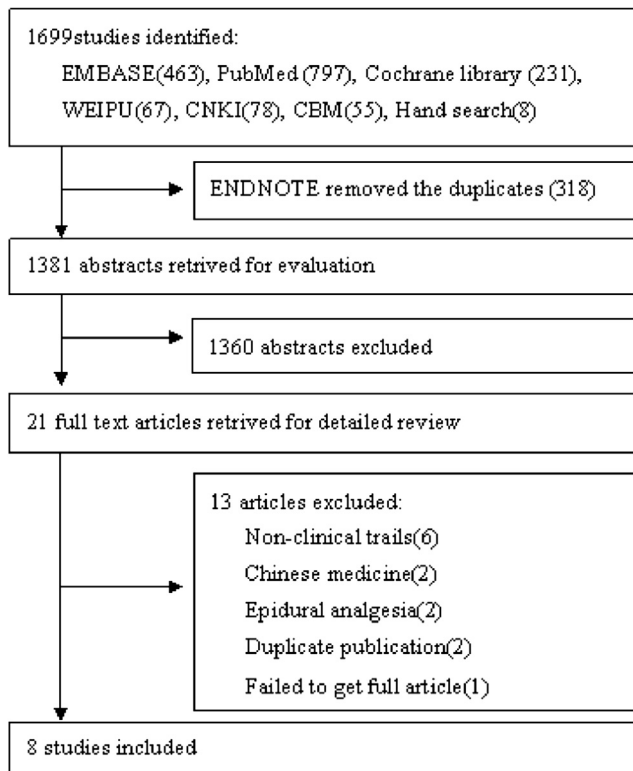


Fig. 1. Flow chart for study selection.

comparable but the procaine group required significant less additional analgesics than placebo group in the other study [25]. The pooled data showed that there was no significant difference in this regard, Relative Risk (RR), 95% Confidence interval (CI): 0.97 [0.46, 2.06]. Nausea, hypodynamia and vomiting were the main adverse effects reported. In one study [24], patients in the procaine group

tended to have higher bodily discomfort while the other study [25] suggested that there was no significant difference on the number of adverse events between groups, RR, 95%CI: 0.91 [0.69, 1.20].

### 3.3. Procaine vs. buprenorphine

Only one study [26] compared procaine with buprenorphine. Pain reduction from beginning (measured by VAS) was significant greater in buprenorphine group than that in procaine group: day 1,  $p < 0.0001$ ; day 2,  $p = 0.005$ ; day 3,  $p = 0.08$ . Buprenorphine group also had less demand for additional analgesics, RR, 95%CI: 13.30 [1.93, 91.55]. Three types of side effects, that is nausea, emesis and sedation, were reported and there was no significant difference on specific side effects or total adverse events, RR, 95%CI: 0.45 [0.18, 1.17]. The study also reported the average length of hospital stay, and there was no significant difference.

### 3.4. Procaine vs. pentazocine

One study [27] compared procaine with pentazocine. Compared with pentazocine group, procaine group had higher pain severity during the first three days: day 1, Mean Difference (MD), 95%CI: 40.0 [35.3, 44.7]; day 2, MD, 95%CI: 24.00 [20.88, 27.12]; day 3, MD, 95%CI: 5.00 [2.17, 7.83]. Besides, pentazocine group had lower requirements for additional analgesics, RR, 95%CI: 2.23 [1.63, 3.05]. Side effects were not reported in this study.

### 3.5. Buprenorphine vs. pethidine

One study [28] compared buprenorphine with pethidine. Two different pain assessment tools, the VAS scale and the 3-mm ordered categories scale, were applied to assess the pain severity; the result showed no significant difference. The needs for additional analgesics and the total number of side effects (nausea, vomiting) were not statistically significant.

Table 1

The main character of included studies.

| Study                   | Country | Age (year)                                                  | Sex (M/F)            | Weight (kg)                                               | Follow up | Severity; (APACHE II-score)                             | Intervention                                                                                                                                             |
|-------------------------|---------|-------------------------------------------------------------|----------------------|-----------------------------------------------------------|-----------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Wilms et al. [24] 2010  | Germany | EG: $50.1 \pm 10.6$ ; CG: $58.4 \pm 14.2$ (mean $\pm$ SD)   | EG: 10/11; CG: 12/9  | EG: $77.3 \pm 19.0$ ; CG: $76.2 \pm 16.4$ (mean $\pm$ SD) | 3 days    | EG: $16.2 \pm 3.2$ ; CG: $15.7 \pm 3.8$ (mean $\pm$ SD) | EG ( $n = 21$ ): Procaine (1.92 g/d); CG ( $n = 21$ ): Placebo (0.9% NaCl)                                                                               |
| Layer et al. [25] 2011  | Germany | Number of patient who was >55 yr old<br>EG: 12; CG: 10      | EG: 16/7; CG: 10/11  | NA                                                        | 3 days    | NA                                                      | EG ( $n = 23$ ): Procaine (2 g/d); CG ( $n = 21$ ): Placebo                                                                                              |
| Jakobs et al. [26] 2000 | Germany | EG: $51.5(26-76)$ ; CG: $47.5(23-72)$ (median, range)       | EG: 12/8; CG: 11/8   | EG: $73(55-93)$ ; CG: $67(48-88)$ (median, range)         | 3 days    | EG: $5(0-16)$ ; CG: $75(50-100)$ (median, range)        | EG ( $n = 20$ ): Procaine, (2 g/d); CG ( $n = 19$ ): Buprenorphine (0.3 mg, initially + 2.4 mg/d)                                                        |
| Stefan et al. [27] 2004 | Germany | EG: $47 \pm 14$ ; CG: $43 \pm 11$ (mean $\pm$ SD)           | EG: 34/17; CG: 38/12 | EG: $73 \pm 14$ ; CG: $77 \pm 17$ (mean $\pm$ SD)         | 4 days    | EG: $5 \pm 4$ ; CG: $4 \pm 3$ (mean $\pm$ SD)           | EG ( $n = 51$ ): Procaine (2 g/d); CG ( $n = 50$ ): Pentazocine (30 mg/6 h)                                                                              |
| Blamey et al. [28] 1984 | UK      | NA                                                          | NA                   | NA                                                        | 1 day     | NA                                                      | EG ( $n = 17$ ): Buprenorphine (0.3 mg); CG ( $n = 15$ ): Pethidine (100 mg)                                                                             |
| Marcia et al. [29] 2002 | USA     | 26–47 (range)                                               | 18/14                | EG: $169(132-244)$ ; CG: $168(108-263)$ (median, range)   | 3 days    | NA                                                      | EG ( $n = 15$ ): Fentanyl (50 mcg/hr) + pethidine (50–100 mg, IM/3 h); CG ( $n = 17$ ): Placebo + pethidine (50–100 mg, IM/3 h)                          |
| He et al. [30] 2006     | China   | 23–56 (range)                                               | 14/6                 | 42–70 (range)                                             | 3 days    | NA                                                      | EG ( $n = 10$ ): Fentanyl (1.5 mg) + Atropine (1.5 mg) + droperidol (10 mg) + lidocaine (300 mg); CG ( $n = 10$ ): Pethidine (50 mg) + Atropine (0.5 mg) |
| Peiró et al. [31] 2008  | Spain   | EG: $54.48 \pm 13.5$ ; CG: $55.18 \pm 18.8$ (mean $\pm$ SD) | EG: 3/5; CG: 5/3     | NA                                                        | 1 day     | NA                                                      | EG ( $n = 8$ ): Metamizole, 2 g/8 h; CG ( $n = 8$ ): Morphine, 1%, 10 mg/4 h                                                                             |

EG: Experimental Group; CG: Control Group; NA: not available; I.M.: injectio intramusculosa; d: day; h: hour.

**Table 2**

The methodology quality of included studies.

| Study                  | Adequate sequence generation | Adequate concealment of allocation | Adequate blinding | Incomplete outcome data | Selective reporting | Other sources of bias |
|------------------------|------------------------------|------------------------------------|-------------------|-------------------------|---------------------|-----------------------|
| Wilms et al [24] 2010  | Unclear                      | Yes                                | Yes               | Yes                     | Unclear             | Unclear               |
| Layer et al [25] 2011  | Unclear                      | Yes                                | Yes               | Yes                     | Unclear             | Unclear               |
| Jakobs et al [26] 2000 | Unclear                      | Unclear                            | No                | No                      | Unclear             | Unclear               |
| Stefan et al [27] 2004 | Unclear                      | Unclear                            | No                | No                      | Unclear             | Unclear               |
| Blamey et al [28] 1984 | Yes                          | Yes                                | Unclear           | Yes                     | Unclear             | Unclear               |
| Marcia et al [29] 2002 | Yes                          | Yes                                | Yes               | No                      | Unclear             | Unclear               |
| He et al [30] 2006     | Unclear                      | Unclear                            | Unclear           | Yes                     | Unclear             | Unclear               |
| Peiró et al [31] 2008  | Unclear                      | Yes                                | No                | Yes                     | Unclear             | Unclear               |

### 3.6. Fentanyl + pethidine vs. placebo + pethidine

One study [29] compared fentanyl + pethidine with placebo + pethidine. The study suggested that the fentanyl group had significantly less pain on admission ( $p < 0.05$ , measured by VAS) and shorter hospitalization ( $p < 0.05$ ) than the placebo group. The study also considered self-reported satisfaction of pain management as an outcome and there was no significant difference.

**Table 3**

Summary of main findings.

| Experimental/control                                                                 | Experimental group (Mean $\pm$ SD or Events/total) | Control group (Mean $\pm$ SD or Events/Total) | Estimated effect between experimental and control group | Participants (number of studies) | Grade <sup>b</sup> |
|--------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------|---------------------------------------------------------|----------------------------------|--------------------|
| <b>Pain severity-day 1</b>                                                           |                                                    |                                               |                                                         |                                  |                    |
| Procaine/placebo [24] <sup>d</sup>                                                   | 5.1 $\pm$ 2.8                                      | 3.9 $\pm$ 2.7                                 | MD,95%CI: 1.20 [−0.46, 2.86]                            | 42(1)                            | Low                |
| Procaine/pentazocine [27] <sup>c</sup>                                               | 75 $\pm$ 12                                        | 35 $\pm$ 12                                   | MD,95%CI: 40.0[35.3,44.7]                               | 101(1)                           | Very low           |
| Buprenorphine/pethidine [28] <sup>c</sup>                                            | 9.1 <sup>f</sup>                                   | 8.4 <sup>f</sup>                              | Not significant <sup>a</sup>                            | 32(1)                            | Moderate           |
| Fentanyl + atropine + droperidol + lidocaine/meperidine + atropine [30] <sup>c</sup> | 1.34 $\pm$ 0.18                                    | 6.8 $\pm$ 2.4                                 | MD,95%CI: −5.46[−6.95, −3.97]                           | 20(1)                            | Low                |
| Metamizole/morphine [31] <sup>c</sup>                                                | 0.2 $\pm$ 0.3                                      | 2.8 $\pm$ 0.4                                 | MD,95%CI: −2.60[−2.95, −2.25]                           | 16(1)                            | Moderate           |
| <b>Pain severity-day 2</b>                                                           |                                                    |                                               |                                                         |                                  |                    |
| Procaine/placebo [24] <sup>d</sup>                                                   | 4.0 $\pm$ 3.1                                      | 2.9 $\pm$ 3.1                                 | MD,95%CI: 1.10[−0.75,2.95]                              | 42(1)                            | Low                |
| Procaine/pentazocine [27] <sup>c</sup>                                               | 39 $\pm$ 8                                         | 15 $\pm$ 8                                    | MD,95%CI: 24.00[20.88,27.12]                            | 101(1)                           | Very low           |
| Fentanyl + atropine + droperidol + lidocaine/meperidine + atropine [30] <sup>c</sup> | 1.04 $\pm$ 0.12                                    | 6.82 $\pm$ 2.6                                | MD,95%CI: −5.78 [−7.39, −4.17]                          | 20(1)                            | Low                |
| <b>Pain severity-day 3</b>                                                           |                                                    |                                               |                                                         |                                  |                    |
| Procaine/placebo [24] <sup>d</sup>                                                   | 5.1 $\pm$ 2.8                                      | 5.1 $\pm$ 2.8                                 | MD,95%CI: 0.00 [−1.69, 1.69]                            | 42(1)                            | Low                |
| Procaine/pentazocine [27] <sup>c</sup>                                               | 15 $\pm$ 9                                         | 10 $\pm$ 5                                    | MD,95%CI: 5.00[2.17,7.83]                               | 101(1)                           | Very low           |
| <b>Pain reduction-day 1</b>                                                          |                                                    |                                               |                                                         |                                  |                    |
| Procaine/placebo [25] <sup>c</sup>                                                   | −27,(−77, 34) <sup>e</sup>                         | −33,(−85, 18) <sup>e</sup>                    | Not significant <sup>a</sup>                            | 44(1)                            | Moderate           |
| Procaine/buprenorphine [26] <sup>c</sup>                                             | 25 <sup>f</sup>                                    | 55 <sup>f</sup>                               | $P < 0.0001^a$                                          | 39(1)                            | Very low           |
| <b>Pain reduction-day 2</b>                                                          |                                                    |                                               |                                                         |                                  |                    |
| Procaine/placebo [25] <sup>c</sup>                                                   | −40,(−82, 12) <sup>e</sup>                         | −50,(−95, 5) <sup>e</sup>                     | Not significant <sup>a</sup>                            | 44(1)                            | Moderate           |
| Procaine/buprenorphine [26] <sup>c</sup>                                             | 40 <sup>f</sup>                                    | 62 <sup>f</sup>                               | $P = 0.005^a$                                           | 39(1)                            | Very low           |
| <b>Pain reduction-day 3</b>                                                          |                                                    |                                               |                                                         |                                  |                    |
| Procaine/placebo [25] <sup>c</sup>                                                   | −39,(−89, 32) <sup>e</sup>                         | −62,(−95, 5) <sup>e</sup>                     | $P = 0.025^a$                                           | 44(1)                            | Moderate           |
| Procaine/buprenorphine [26] <sup>c</sup>                                             | 52.2 <sup>f</sup>                                  | 70 <sup>f</sup>                               | $P = 0.08^a$                                            | 39(1)                            | Very low           |
| <b>Need for additional analgesics</b>                                                |                                                    |                                               |                                                         |                                  |                    |
| Procaine/placebo [24,25]                                                             | 10/44                                              | 10/42                                         | RR ,95%CI: 0.97 [0.46, 2.06]                            | 86(2)                            | Moderate           |
| Procaine/buprenorphine [26]                                                          | 14/20                                              | 1/19                                          | RR ,95%CI: 13.30 [1.93, 91.55]                          | 39(1)                            | Very low           |
| Buprenorphine/pethidine [28]                                                         | 13/17                                              | 14/15                                         | RR ,95%CI: 0.82 [0.61, 1.10]                            | 32(1)                            | Moderate           |
| Procaine/pentazocine [27]                                                            | 50/51                                              | 22/50                                         | RR,95%CI: 2.23 [1.63, 3.05]                             | 101(1)                           | Very low           |
| Metamizole/morphine [31]                                                             | 3/8                                                | 3/8                                           | RR, 95%CI:1.00 [0.28, 3.54]                             | 16(1)                            | Moderate           |
| <b>Side effects</b>                                                                  |                                                    |                                               |                                                         |                                  |                    |
| Procaine/placebo [25]                                                                | 18/23                                              | 18/21                                         | RR ,95%CI: 0.91 [0.69, 1.20]                            | 43(1)                            | Moderate           |
| Procaine/buprenorphine [26]                                                          | 5/60                                               | 11/57                                         | RR ,95%CI: 0.45 [0.18, 1.17]                            | 39(1)                            | Very low           |
| Buprenorphine/pethidine [28]                                                         | 15/17                                              | 11/15                                         | RR ,95%CI: 1.20 [0.85, 1.71]                            | 32(1)                            | Moderate           |
| Fentanyl + atropine + droperidol + lidocaine/meperidine + atropine [30]              | 3/10                                               | 10/10                                         | RR ,95%CI: 0.33 [0.14, 0.80]                            | 20(1)                            | Low                |
| Metamizole/morphine [31]                                                             | 1/8                                                | 4/8                                           | RR ,95%CI: 0.33 [0.06, 1.83]                            | 16(1)                            | Moderate           |

Note: pain severity were measured by visual analog scale in most study except study carried out by Stefan et al [27] which was measured by a 9-point scale questionnaire.

<sup>a</sup> Data were abstracted directly from original studies.

<sup>b</sup> Quality of evidence are showed as: High quality: Further research is very unlikely to change our confidence in the estimate of effect; Moderate quality: Further research is likely to have an important impact on our confidence in the estimate of effect and may change the estimate; Low quality: Further research is very likely to have an important impact on our confidence in the estimate of effect and is likely to change the estimate; Very low quality: We are very uncertain about the estimate.

<sup>c</sup> Pain measured by visual analog scale.

<sup>d</sup> Pain measured by a 9-point semiquantitative scale.

<sup>e</sup> Effect presented as mean, range.

<sup>f</sup> Effect presented as median.

### 3.7. Fentanyl + atropine + droperidol + lidocaine vs. pethidine + atropine

One study [30] compared the combination of fentanyl, atropine, droperidol and lidocaine with the combination of pethidine and atropine for pain relief in acute pancreatitis. The combination of fentanyl, atropine, droperidol and lidocaine inclined to lead to lower VAS score: day 1, MD, 95%CI: −5.46 [−6.95, −3.97]; day 2, MD, 95%CI: −5.78 [−7.39, −4.17]. Again, it also associated with lower

sedation score. The number of vomiting in each group was reported and the pethidine-atropine group had significant higher vomiting rate RR 95%CI 0.33 [0.14, 0.80].

### 3.8. Morphine vs. metamizole

One study [31] compared morphine with metamizole. Patients treated with metamizole tended to have lower VAS score than those treated with morphine, MD, 95%CI:  $-2.60 [-2.95, -2.25]$ . Besides, there was no significant difference between the two groups regarding the demand for a rescue treatment with pethidine, RR, 95%CI: 1.00 [0.28, 3.54]. The number of patients responded in the first 24 h of hospitalization in two groups was comparable, RR, 95%CI: 2.00 [0.75, 5.33]. There was also no difference in the mean time to achieve pain relief, MD, 95%CI:  $-7.00 [-20.48, 6.48]$ , and the total number of side effects, OR, 95%CI 0.33 [0.06, 1.83].

## 4. Discussion

In this study, we failed to find out the most effective analgesics for pain relief in acute pancreatitis because the available evidence was limited and the overall quality of evidence was low. However, buprenorphine, pentazocine and metamizole showed relatively better efficacy. For every comparison in this systematic review, the safety data was comparable between experimental and control group, except that the combination of pethidine and atropine led to a significant higher vomiting rate than the combination of fentanyl, atropine, droperidol and lidocaine.

Application of procaine in patients with acute pancreatitis is mainly adopted in the German-speaking world and China [32–34]. Included studies in this review used 1.9–2.0 mg procaine to relieve pain for patients. The results of this systematic review indicate that procaine is ineffective for pain relief in acute pancreatitis. Though one study [25] suggested that procaine was associated with a stronger decrease in pain and less needs for additional analgesics, another study [24] showed a completely opposite result. The pooled data of the two studies suggested that there was no significant difference in the need for additional analgesics. Moreover, Kahl et al. [27] confirmed that procaine was ineffective in reducing pain in acute pancreatitis and might even lead to higher additional needs for analgesics.

Compared with procaine, buprenorphine and pentazocine are superior in efficiency and comparable in safety. Buprenorphine is a semisynthetic opioid and acts as a partial agonist at the  $\mu$ -receptor and  $\kappa$ -receptor. Past studies have demonstrated that it is effective and safe for the management of postoperative pain [35,36]. An early experimental study [37] suggested that buprenorphine could distinctly reduce pain sensitivity in acute pancreatitis. Buprenorphine was also recommended by some guidelines [11,38,39]. Pentazocine is an agonist-antagonist analgesics in the benzomorphan class of opioids. Under the therapeutic dosage, pentazocine is a potent analgesic with favorable safety, and it is well tolerated and unlikely to induce respiratory depression [40].

Traditionally, morphine should be avoided to use for patients with acute pancreatitis as it may cause spasm in the sphincter of Oddi [38,41]. According to a small scale study [31] included in this systematic review, morphine tended to be less effective in pain management than metamizole and their safety were comparable. Pethidine is prescribed instead of morphine to avoid inducing sphincter of Oddi spasm. Our study shows that pethidine is comparable to relieve pain as buprenorphine but less effective compared with the combination of fentanyl and pethidine. The combination of pethidine and atropine is inclined to lead to higher pain severity and vomiting rate than the combination of fentanyl, atropine, droperidol

and lidocaine. Since normeperidine, a toxic metabolite of pethidine, may increase the risk of neuromuscular irritation, seizures, myoclonus and tremors, specific concern must be taken especially for patients with renal failure or alcoholic pancreatitis [42]. For safety concern, A.M. Peiró et al. recommended morphine and pethidine as the second-line analgesic drug for acute pancreatitis.

Fentanyl shows a favorable effect on pain management in acute pancreatitis when combined with other medicaments. Apart from the fact that the combination of fentanyl and pethidine shows a better pain relief effect compared with single pethidine, the combination of fentanyl, atropine, droperidol and lidocaine is also more effective than the combination of pethidine and atropine. However, fentanyl may also cause spasm of the sphincter of Oddi [43], but the incidence is low under small dose [44]. What is worse, the steady-state blood level of fentanyl is slow to be achieved, which may lead to a higher requirement for additional analgesics [45,46].

According to our search, there are only two trials [31,47] that used NSAIDs in patients with acute pancreatitis. One study [31] suggests that intravenous metamizole showed a non-significant association with a quicker pain relief than morphine in patients with acute pancreatitis. The other study [47], which we failed to get the full article, indicates that indomethacin may lead to shorter-time pain and less number of opiate injections than placebo. What is more, no bleeding from the gastrointestinal tract was observed in the indomethacin group. Overall, NSAIDs show a favorable efficacy and safety for pain control in acute pancreatitis and this has been approved by a recent published systematic review [48].

The conclusion of this systematic review is limited by the quality of the included trials. Five studies [24–27,30,31] did not report the method for randomization, and the concealment of randomization was not clear in three studies [26,27,30]. Besides, three studies [26,27,31] were open and three studies [26,27,29] have high possibility that the data given was incomplete. The risk of bias and imprecision of results (the number of patients of all studies was small) were the main reasons for the downgradation of the quality of evidence and we hardly found any possible factors that may upgrade the level of evidence.

Pain alleviation is a vital step in the management of acute pancreatitis and many guidelines suggest using narcotic agents to relieve it [11,17,38]. However no guideline provided the most recommended agent due the limitation in current evidence. The dosage of narcotic agent and the frequency of administration should be monitored closely by experienced physicians. There are several limitations in our study. Firstly, most participants of the studies were white people from Europe, which lead to a concern that if our study results can be applied on other ethnic group. Secondly, the numbers of patients and studies on every comparison were small which greatly restrained the quality of evidence. Lastly, we did not assess the risk of publication bias as the total number of included studies was small.

Overall, the current evidences do not suggest routinely using procaine for pain relief for acute pancreatitis. Pethidine and fentanyl are effective but special attention must be taken in order to avoid side effects. Buprenorphine, pentazocine and NSAIDs show relatively better efficiency and safety. However, more studies are still wanted to confirm the efficacy of these agents due to the limitations of current evidence. RCTs comparing different analgesics were of low quality and did not clearly favor any particular analgesic for pain relief in acute pancreatitis.

## Author contributions

Wenbo Meng, Xun Li and Jinqu Yuan contributed to the selection of studies and data extraction. All authors contributed to the study design, data analysis, and interpretation of results, and

reviewed the manuscript for important intellectual content and approved the final version.

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